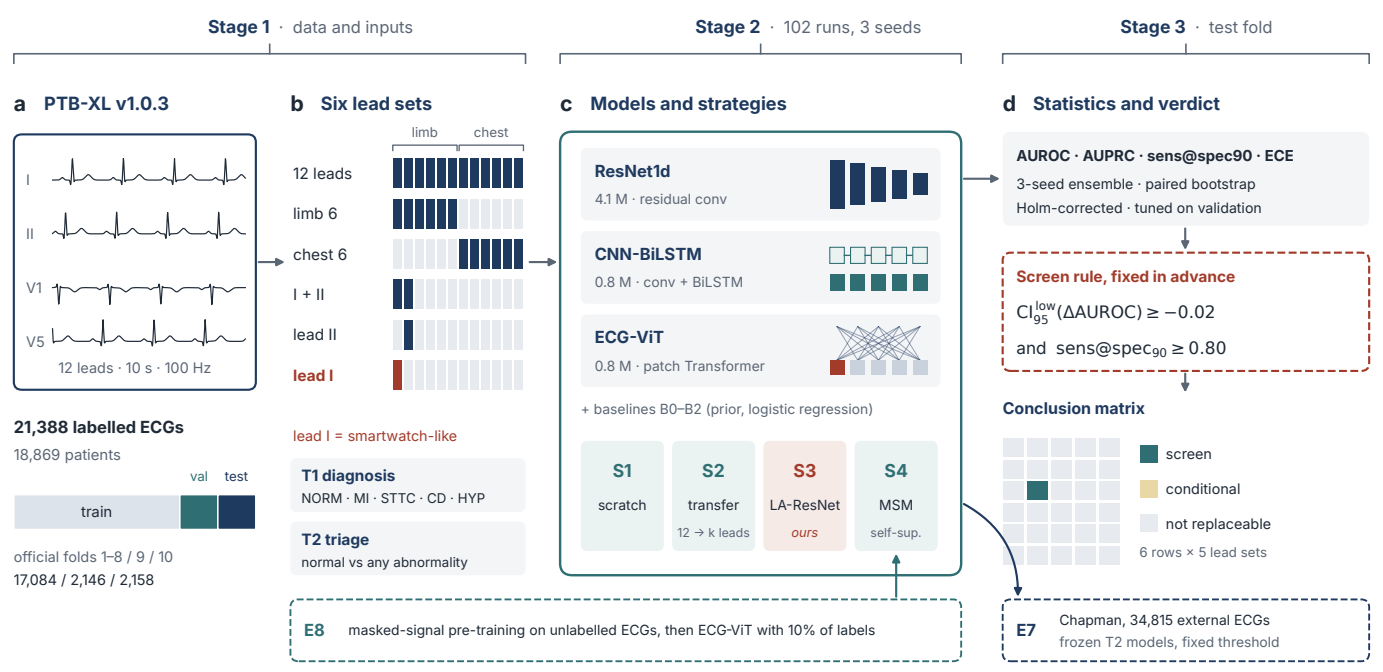
Stage 1 · data and inputs
Stage 2 · 102 runs, 3 seeds
Stage 3 · test fold
a PTB-XL v1.0.3
I
II
V1
V5
12 leads · 10 s · 100 Hz
21,388 labelled ECGs
18,869 patients
val
test
train
official folds 1–8 / 9 / 10
17,084 / 2,146 / 2,158
b Six lead sets
limb
chest
12 leads
limb 6
chest 6
I + II
lead II
lead I
lead I = smartwatch-like
T1 diagnosis
NORM · MI · STTC · CD · HYP
T2 triage
normal vs any abnormality
c Models and strategies
ResNet1d
4.1 M · residual conv
CNN-BiLSTM
0.8 M · conv + BiLSTM
ECG-ViT
0.8 M · patch Transformer
+ baselines B0–B2 (prior, logistic regression)
S1
scratch
S2
transfer
12 → k leads
S3
LA-ResNet
ours
S4
MSM
self-sup.
E8
masked-signal pre-training on unlabelled ECGs, then ECG-ViT with 10% of labels
d Statistics and verdict
AUROC · AUPRC · sens@spec90 · ECE
3-seed ensemble · paired bootstrap
Holm-corrected · tuned on validation
Screen rule, fixed in advance
$CI_{95}^{low}(\Delta AUROC) \geq -0.02$
and $sens@spec_{90} \geq 0.80$
Conclusion matrix
screen
conditional
not replaceable
6 rows × 5 lead sets
E7
Chapman, 34,815 external ECGs
frozen T2 models, fixed threshold